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Chapter 1

Results — Forecasting Models

- Six runs: LEAR / DNN / Transformer × DK1 / DE, identical hourly test window, compared per market on their common surviving test days (DK1: 187; DE: 176 — the Transformer's 14-day input sequence drops more leading days). Setup and deviations: Ch. **??**.

1.1 Headline accuracy

Table 1.1: DK1: final benchmark metrics (test 2025-10-24–2026-05-31, common days).

	MAE	RMSE	sMAPE	rMAE
LEAR	17.574	26.891	29.048	0.469
DNN	17.960	25.634	29.339	0.482
Transformer	20.076	28.909	31.767	0.543
Naive (weekly)	36.899	51.385	50.110	1.000

Table 1.2: DE: final benchmark metrics (test 2025-10-24–2026-05-31, common days).

	MAE	RMSE	sMAPE	rMAE
LEAR	13.069	25.041	22.107	0.376
DNN	13.476	22.544	22.763	0.388
Transformer	14.719	23.755	23.901	0.420
Naive (weekly)	35.049	51.914	46.090	1.000

- rMAE ranking on BOTH markets: LEAR < DNN < Transformer (LEAR best). All three beat the weekly-naive (rMAE < 1).
- RMSE: the DNN edges LEAR on RMSE (DK1 25.6 vs 26.9; DE 22.5 vs 25.0) — it trims the largest errors — but rMAE/MAE (the paper's primary metric, [1] §5.4) favour LEAR.

1.2 Statistical significance

- Read cell (r, c) as the p-value that column c is more accurate than row r ([1] Fig. 5 convention); bold/threshold at 0.05.
- **LEAR vs DNN: not significant** on either market (DK1 $p = 0.16$, DE $p = 0.19$ for LEAR-beats-DNN). LEAR is numerically ahead but the two are statistically indistinguishable — mirroring [1]'s finding that the DNN/LEAR gap is frequently insignificant.

Table 1.3: DK1: GW multivariate p-values (column model better than row model).

	LEAR	DNN	Transformer	Naive (weekly)
LEAR	–	0.836	1.000	1.000
DNN	0.164	–	1.000	1.000
Transformer	0.001	0.001	–	1.000
Naive (weekly)	0.000	0.000	0.000	–

Table 1.4: DE: GW multivariate p-values (column model better than row model).

	LEAR	DNN	Transformer	Naive (weekly)
LEAR	–	0.807	0.988	1.000
DNN	0.193	–	0.994	1.000
Transformer	0.012	0.006	–	1.000
Naive (weekly)	0.000	0.000	0.000	–

- **Transformer significantly worse** than both LEAR and DNN on both markets ($p \leq 0.012$).
- All models significantly beat the weekly-naive ($p = 0$).

1.3 Computation cost

Table 1.5: Amortised recalibration cost (full-backtest wall-clock / test day). Full timings in `Report/metrics/run_timings.md`.

Model	**DK1 s/day**	**DE s/day**	**Hardware / cadence**
LEAR	50.3	41.3	CPU; one LassoCV per period × 3 windows, daily
DNN	0.9	1.6	GPU; one MIMO net × 4-seed ensemble, daily
Transformer	1.6	1.9	GPU; one MIMO net, weekly refit

- **Hardware-driven cost inversion**: contrary to [1] Table 4 (hourly, CPU: LEAR seconds, DNN minutes), here the GPU MIMO models are *30–50× cheaper* per recalibration than per-period CPU LassoCV. The DNN matches LEAR's accuracy (statistically tied, above) at ≈1/40th the cost — the practical case for the DNN on this platform.

1.4 Comparison with published benchmarks

- Caveat first: test periods differ (papers 2016–2018 vs. this 2025–2026); only rMAE *levels* and *rankings* are comparable, never absolute MAE ([1] §5.4).

- Levels are in the same ballpark as the paper; DK1 slightly higher (worse), DE LEAR actually *below* the paper's EPEX-DE figure.
- **Ranking inverts both source papers**: [1] reports DNN ≥ LEAR and [2] reports Transformer ≥ DNN; here it is LEAR ≥ DNN ≥ Transformer. Likely because the deep models are under-optimised relative to LEAR in this study (DNN: 30-trial

Table 1.6: Our ensemble rMAE vs published ([1] Table 3). Periods differ; rankings/levels only.

	DK1 ours	**NP [1]**	**DE ours**	**EPEX-DE [1]**
LEAR	0.469	0.420	0.376	0.433
DNN	0.482	0.407	0.388	0.374

TPE vs the paper's 1500; Transformer: paper Nord-Pool hyperparameters reused unretuned, adapted into a calibration-window/weekly-refit pipeline rather than its native train-once+renormalise), whereas LEAR re-selects λ by cross-validation at every daily recalibration. Not evidence that the architectures are inherently weaker — see future work.

1.5 Threats to validity (modelling side)

- 31-week test window (vs the papers' 1–2 years); a single run per DNN/Transformer config (seed variance only partly mitigated by the DNN's 4-seed ensemble; the Transformer is a single seed).
- Deep models under-tuned relative to LEAR (see Table 1.6 discussion and the deviations table, Ch. **??**, Table **??**).
- 2025–2026 regime includes negative-price and high-renewables days not present in the papers' data; hourly-downsampled 15-min markets.

1.6 Future work (modelling side)

- Full-budget DNN tuning (toward the paper's 1500 trials) and per-market Transformer hyperparameter search before drawing architectural conclusions.
- Native train-once + renormalise Transformer path ([2] §3.3) instead of calibration-window refit; 1456-day LEAR window once data depth allows; LARS+AIC λ-selection; probabilistic extension (pinball/CRPS); per-hour DM profiles.

Bibliography

[1] Jesus Lago, Grzegorz Marcjasz, Bart De Schutter, and Rafał Weron. Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark. *Applied Energy*, 293:116983, 2021.

[2] Jesús Llorente and José Portela. A transformer approach for electricity price forecasting. *arXiv preprint arXiv:2403.16108*, 2025.